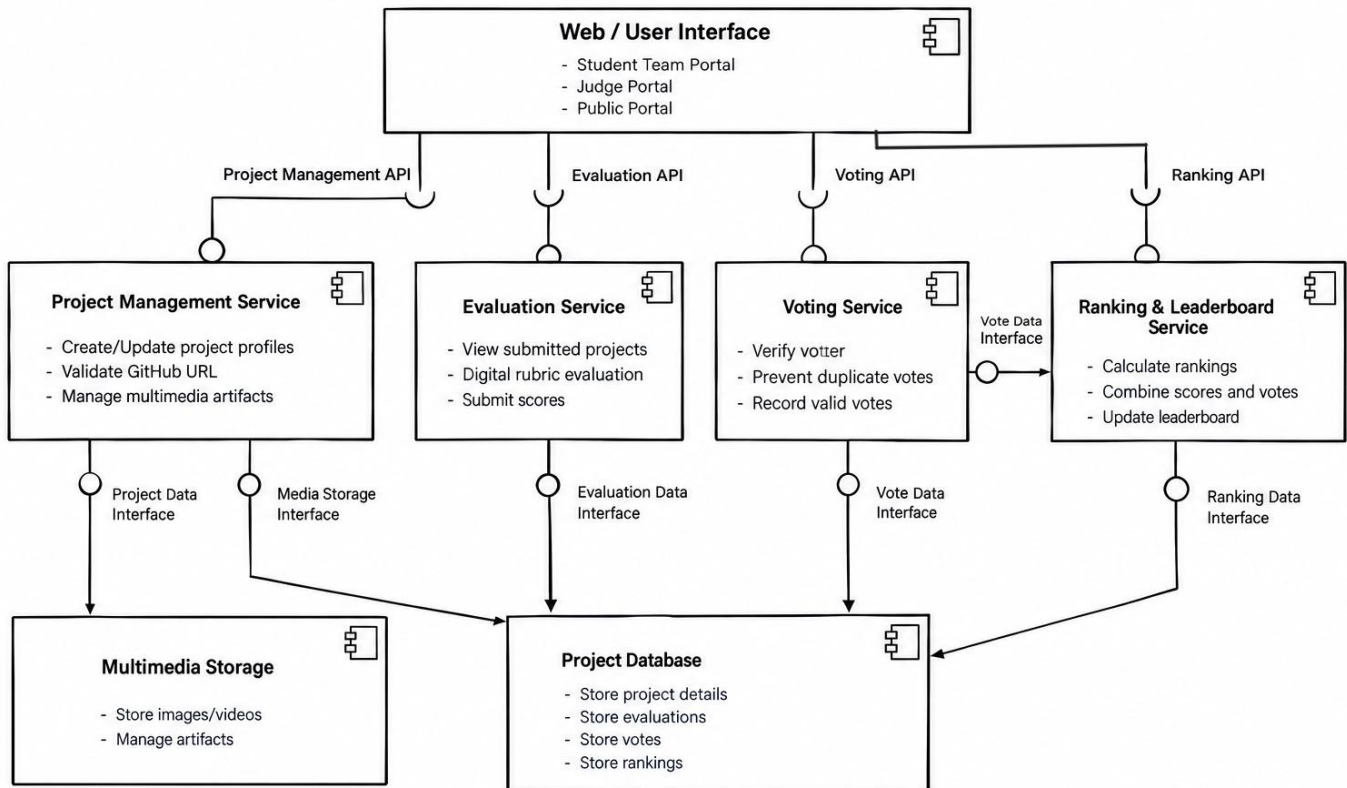


SE LAB 03

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Architecture Selection and Justification

Primary Architecture Style: Component-Based Architecture (CBA)

The Student Project Portfolio & Showcase App follows a Component-Based Architecture, where the system is divided into modular components with clearly defined responsibilities. Each component performs a specific function and communicates with other components through well-defined interfaces. This reduces direct dependency between components and makes the system easier to maintain and modify.

The major components of the system are Web/User Interface, Project Management Service, Evaluation Service, Voting Service, Ranking & Leaderboard Service, Project Database, and Multimedia Storage.

Component Responsibilities

- Web/User Interface**: Provides the interface through which student teams, judges, and public users interact with the system.
- Project Management Service**: Handles project profile creation, project submission, GitHub URL validation, and project artifact management.
- Evaluation Service**: Allows judges to view submitted projects and record scores using digital evaluation rubrics.
- Voting Service**: Handles public voting, voter verification, and duplicate-vote prevention.

- Ranking & Leaderboard Service: Processes valid scores and votes to calculate and display project rankings.
- Project Database: Stores project details, submissions, evaluations, votes, and ranking-related information.
- Multimedia Storage: Stores project videos, images, and other multimedia artifacts.

Interface and Component Interaction

The components communicate using explicit interfaces. The Web/User Interface interacts with the Project Management, Evaluation, Voting, and Ranking services through their respective interfaces. The Project Management and Evaluation services interact with the Project Database to store and retrieve information. The Voting Service provides valid voting information to the Ranking & Leaderboard Service so that rankings can be updated.

The diagram uses UML provided and required interfaces to represent these interactions. The interfaces make the communication between components clear without exposing their internal implementation.

Data Flow

The main data flow through the system is:

Student Project Information → Project Management Service → Project Database

Judge Evaluation → Evaluation Service → Project Database

Public Vote → Voting Service → Ranking & Leaderboard Service

Project Media → Project Management Service → Multimedia Storage

The Ranking & Leaderboard Service uses valid project scores and voting information to calculate the current rankings and make the leaderboard available through the user interface.

Justification for Component-Based Architecture

Component-Based Architecture is suitable for this application because the system contains several distinct functions that can be separated into individual components. Each component has a focused responsibility and communicates through defined interfaces.

This architecture provides modularity, since project management, evaluation, voting, ranking, and storage are separated. It also improves maintainability, because changes to one component can be made with minimal impact on other components.

The architecture also supports security by allowing sensitive operations, particularly voter verification and duplicate-vote prevention, to be handled within the Voting Service. Other components do not need direct access to the internal voting mechanism.

From a performance perspective, separating the major functions allows frequently used operations such as public voting and leaderboard updates to be handled independently. This supports the application's requirement for responsive voting and ranking operations. Therefore, Component-Based Architecture provides a clear, modular, and manageable structure for the Student Project Portfolio & Showcase App while supporting its project submission, evaluation, voting, and ranking requirements.